

Exercise 3 – SQL CASE Statements

EXERCISE 3 - SQL CASE STATEMENTS

```

① SELECT product_name,
      price,
      CASE
        WHEN price > 1000 THEN 'Expensive'
        WHEN price BETWEEN 100 AND 1000
          THEN 'Mid-range'
        ELSE 'Budget'
      END AS price_category
FROM products;

```

product_name	price	price_category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45	Budget
Monitor	300	Mid-range
Mouse	25	Budget

```

② SELECT customer_name,
      amount,
      CASE
        WHEN amount >= 1000 THEN 'High Value'
        WHEN amount BETWEEN 500 AND 999.99
          THEN 'Medium Value'
        ELSE 'Low Value'
      END AS order_value_category
FROM orders;

```

Customer_name	amount	order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
GTHan	1200.00	High Value

EXERCISE 3 - SQL CASE REVIEWER

```

③ SELECT emp_name,
      department,
      salary,
      CASE
        WHEN department = 'IT' AND
             salary > 80000 THEN 'Senior IT'
        WHEN department = 'HR' AND
             salary > 55000 THEN 'Experienced HR'
        ELSE 'Staff'
      END AS position_level
FROM employees;
  
```

emp_name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	staff
Lucey	Finance	95000	staff
Tom	HR	35000	staff

```

④ SELECT student_name,
      score,
      CASE
        WHEN score >= 90 THEN 'A'
        WHEN score BETWEEN 80 AND 89 THEN 'B'
        WHEN score BETWEEN 70 AND 79 THEN 'C'
        WHEN score BETWEEN 60 AND 69 THEN 'D'
        ELSE 'F'
      END AS grade
FROM students;
  
```


student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

```

(5) SELECT delivery-id,
        delivery-time_minutes,
        CASE
            WHEN delivery-time_minutes <= 30
            THEN 'Fast'
            WHEN delivery-time_minutes BETWEEN
                31 AND 60 THEN 'On Time'
            ELSE 'Late' END AS performance
    FROM deliveries;
  
```

delivery-id	delivery-time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

```

(6) SELECT issue_type,
        priority,
        CASE
            WHEN priority = 3 THEN 'High'
            WHEN priority = 2 THEN 'Medium'
            WHEN priority = 1 THEN 'Low'
            ELSE 'Low' END AS priority-label
    FROM tickets;
  
```

issue_type	priority	priority_label
login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

```

(7) SELECT student_id,
      (days-present * 100 / total-days) AS
      attendance_percentage
      CASE
        WHEN attendance_percentage >= 90 THEN
          'Excellent'
        WHEN attendance_percentage BETWEEN
          75 AND 89 THEN 'GOOD'
        WHEN attendance_percentage
        ELSE 'Needs Improvement'
      END AS attendance_status
FROM attendance;

```

Student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

⑧ SELECT product_id,
stock_qty,
CASE
WHEN stock_qty = 0 THEN
'out of Stock'
WHEN stock_qty BETWEEN 1 AND 5
THEN 'Low Stock'
~~WHEN~~
ELSE 'In Stock' END AS stock_status
FROM products_inventory;

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

⑨ SELECT subject,
enrolled_students,
CASE
WHEN enrolled_students >= 25 THEN
'Large'
WHEN enrolled_students BETWEEN 10
AND 24 THEN 'Medium'
ELSE 'Small'
END AS class_size_category
FROM classes;

class_id	Subject		
subject	enrolled_students	class_size_category	
Math	30	Large	
English	25	Large	
Science	15	Medium	
Art	5	Small	
History	20	Medium	

```

10) SELECT payment_id,
      payment_method,
      amount,
      CASE
        WHEN payment_method = 'Cash'
          AND amount >= 200 THEN
          'Eligible for Discount'
        ELSE 'Not Eligible'
      END AS discount_eligibility
FROM payments;

```

payment_id	payment_method	amount	discount_eligibility
1	Card	50	Not Eligible
2	Cash	200	Eligible for discount
3	Card	150	Not Eligible
4	Paypal	75	Not Eligible
5	Cash	300	Eligible for discount